

Combinatorial interpretations in affine Coxeter groups

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- ▶ Geometry: Classification of Lie groups and Lie algebras

Examples of Coxeter groups

A shorthand notation is the **Coxeter graph**:

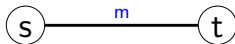
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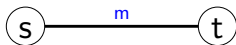


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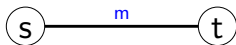
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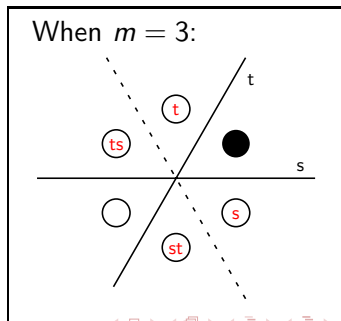
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Symmetry group of regular m -gon.



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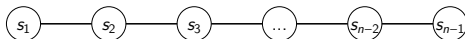
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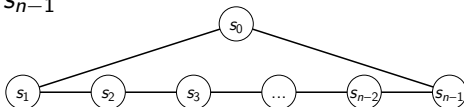
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Affine n -Permutations $\widetilde{S}_n$

- ▶ Generators: $s_0, s_1, \dots, s_{n-1}$
- ▶ Relations:

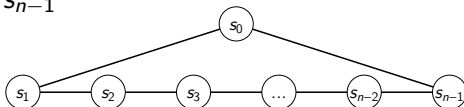


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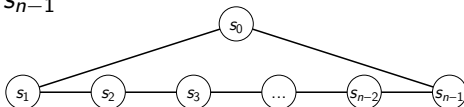


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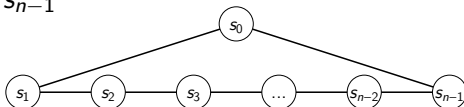


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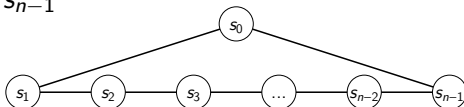


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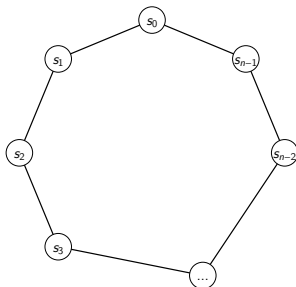
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 - Not quite! (Would add a relation)
- Better to view graph as:
 - Every generator is the same.



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Write an element $\tilde{w} \in \widetilde{S}_n$ in 1-line notation as a **permutation of $\mathbb{Z}$** .

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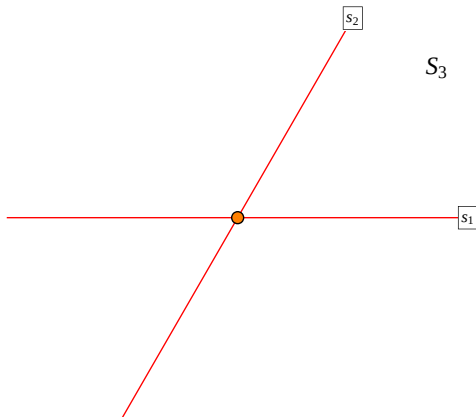
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$\widetilde{w}$ is defined by the **window** $[\widetilde{w}(1), \widetilde{w}(2), \dots, \widetilde{w}(n)]$. $s_1 s_0 = [0, 1, 3, 6]$

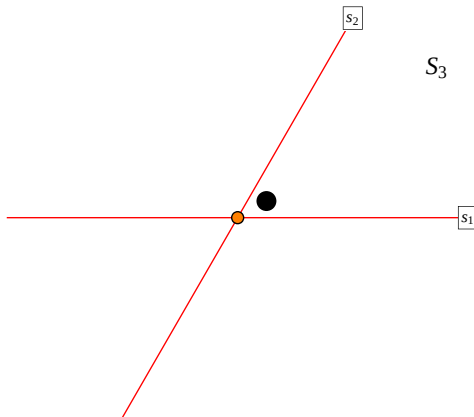
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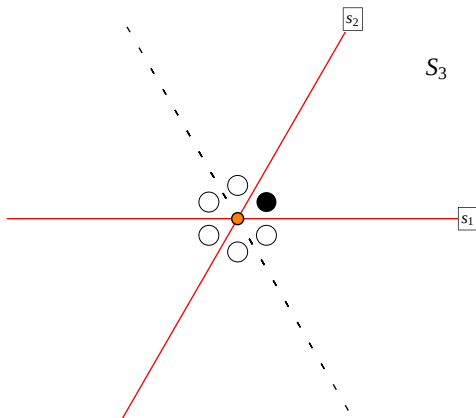
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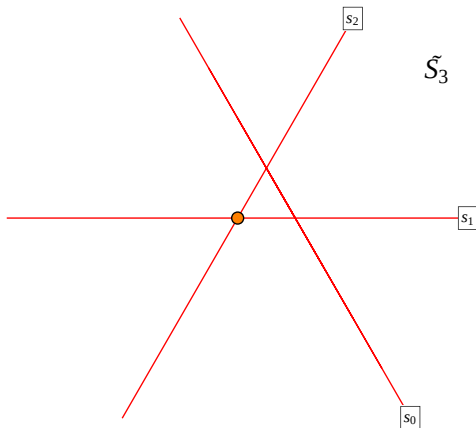
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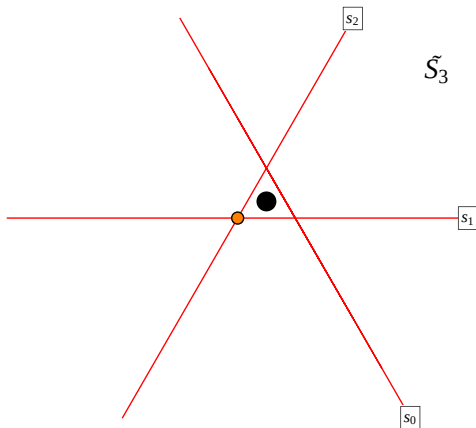
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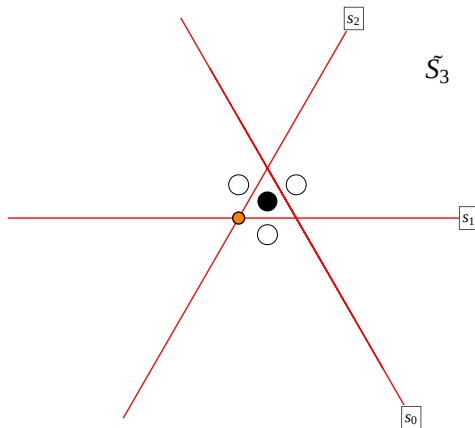
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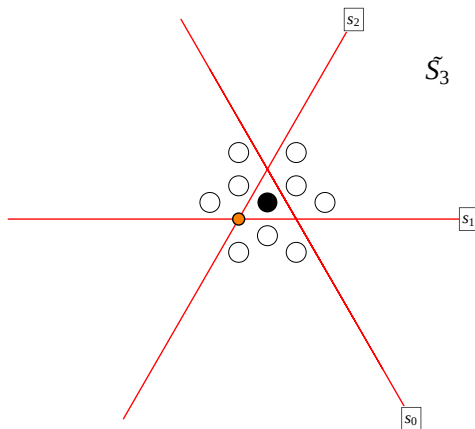
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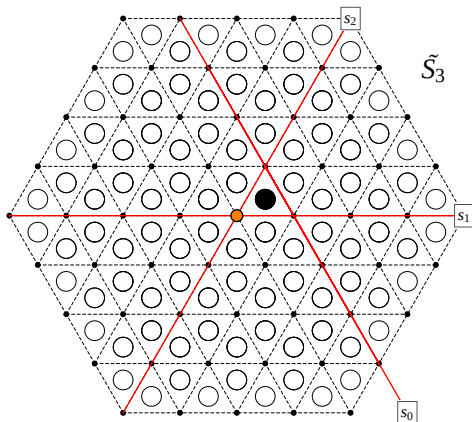
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Affine n -Permutations $\widetilde{S}_n$ — elements correspond to alcoves.



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For a Coxeter group $\widetilde{W}$,

- ▶ An induced subgraph of $\widetilde{W}$'s Coxeter graph is a subgroup W
- ▶ Every element $\widetilde{w} \in \widetilde{W}$ can be written $\widetilde{w} = w^0 w$, where $w^0 \in \widetilde{W}/W$ is a coset representative and $w \in W$.

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 - ▶ w^0 determines the entries; w determines their order.

Example. For $\widetilde{w} = [-11, 20, -3, 4, 11, 0] \in \widetilde{S}_6$,

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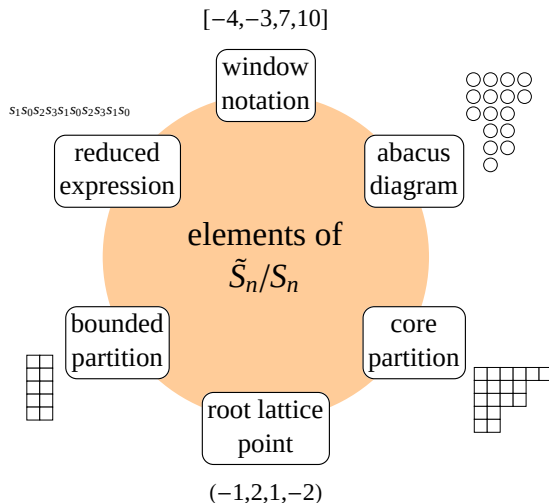
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Many interpretations of these *minimal length coset representatives*.

Combinatorial interpretations of $\tilde{S}_n/S_n$



An abacus model for $\widetilde{S}_n/S_n$

(James and Kerber, 1981) Given $w^0 = [w_1, \dots, w_n] \in \widetilde{S}_n/S_n$,

- Place integers in n *runners*.

−15 −14 −13 −12

−11 −10 −9 −8

−7 −6 −5 −4

−3 −2 −1 0

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

17 18 19 20

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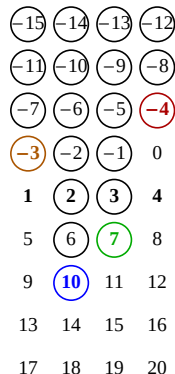
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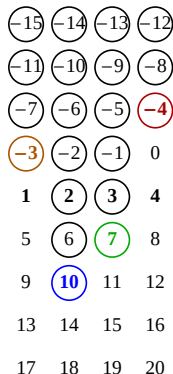
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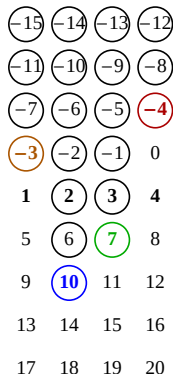
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The generators act nicely on the abacus.



Action of generators on the abacus

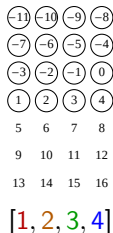
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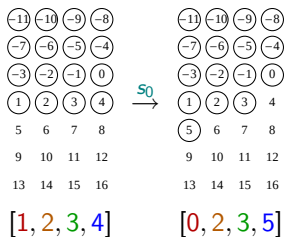


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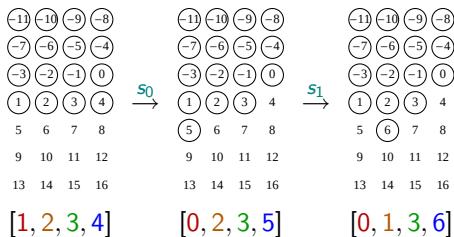


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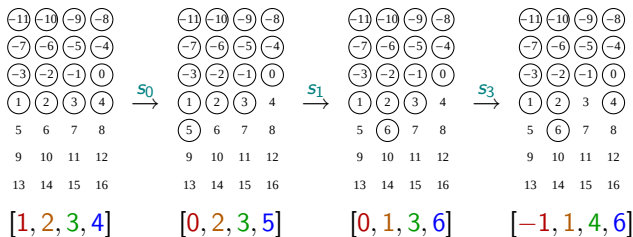


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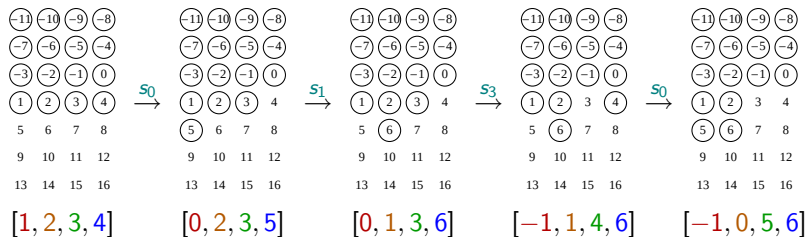


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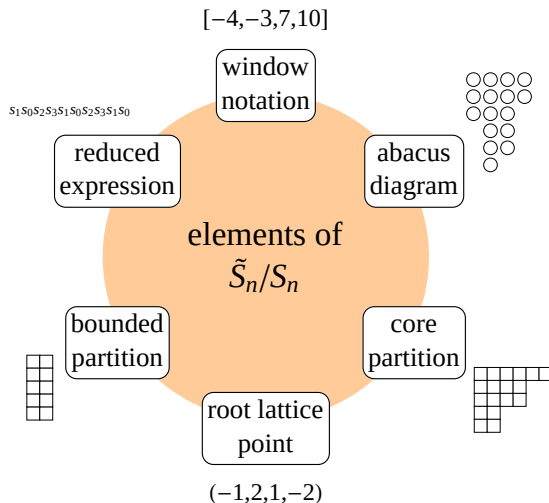
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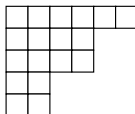


Combinatorial interpretations of $\tilde{S}_n/S_n$

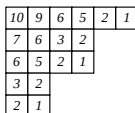


Integer partitions and n -core partitions

For an integer partition $\lambda = (\lambda_1, \dots, \lambda_k)$ drawn as a Ferrers diagram,

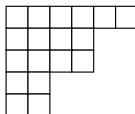


The *hook length* of a box is # boxes below and to the right.

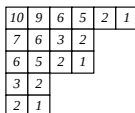


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The *hook length* of a box is # boxes below and to the right.



An n -core is a partition with no boxes of hook length dividing n .

Example. λ is a 4-core, 8-core, 11-core, 12-core, etc.

λ is NOT a 1-, 2-, 3-, 5-, 6-, 7-, 9-, or 10-core.

Core partitions for $\widetilde{S}_n/S_n$

Elements of $\widetilde{S}_n/S_n$ are in bijection with n -cores.

Bijection: $\{\text{abaci}\} \longleftrightarrow \{n\text{-cores}\}$

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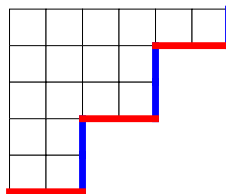
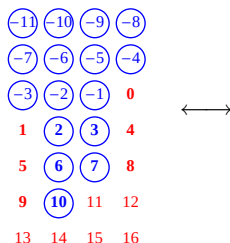
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Rule: Read the boundary steps of λ from the abacus:

► A bead $\leftrightarrow$ vertical step

► A gap $\leftrightarrow$ horizontal step



Action of generators on the core partition

- ▶ Label the boxes of λ with residues.
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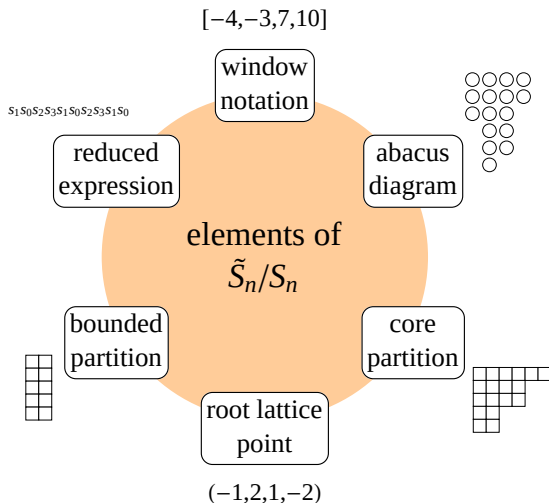
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Combinatorial interpretations of $\tilde{S}_n/S_n$



Bounded partitions for $\tilde{S}_n/S_n$

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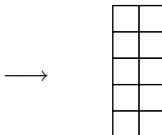
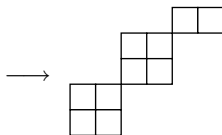
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10	9	6	5	2	1
7	6	3	2		
6	5	2	1		
3	2				
2	1				

$$\lambda = (6, 4, 4, 2, 2)$$



$$\beta = (2, 2, 2, 2, 2)$$

Canonical reduced expression for $\tilde{S}_n/S_n$

Given the bounded partition, read off the reduced expression:

Method: (Berg, Jones, Vazirani, 2009)

- ▶ Fill β with residues i
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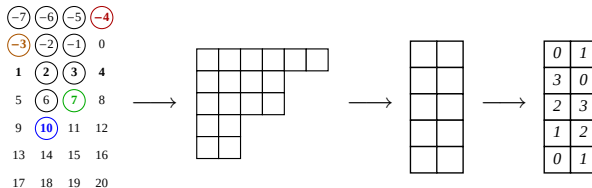
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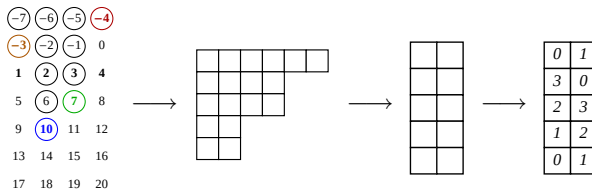
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Definition. An element in a Coxeter group is **fully commutative** if it has only one reduced expression (up to commutation relations).

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Answer: **321**456...

Enumerating fully commutative elements

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Answer: Catalan many!

S_1 : 1. id

S_2 : 2. id, s_1

S_3 : 5. id, s_1 , s_2 , s_1s_2 , s_2s_1

S_4 : 14. id, s_1 , s_2 , s_3 , s_1s_2 , s_2s_1 , s_2s_3 , s_3s_2 , s_1s_3 ,
 $s_1s_2s_3$, $s_1s_3s_2$, $s_2s_1s_3$, $s_3s_2s_1$, $s_2s_1s_3s_2$

Key idea: (Billey, Jockusch, Stanley, 1993)

w is fully commutative $\iff w$ is 321-avoiding.

(Knuth, 1973) These are counted by the Catalan numbers.

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Answer: Infinitely many! (Even in $\widetilde{S}_3$.)

$\text{id}, s_1, s_1 s_2, s_1 s_2 s_0, s_1 s_2 s_0 s_1, s_1 s_2 s_0 s_1 s_2, \dots$

Multiplying the generators cyclically does not introduce braids.

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Question: **How many** fully commutative elements are there in $\widetilde{S}_n$?

Answer: Infinitely many! (Even in $\widetilde{S}_3$.)

$\text{id}, s_1, s_1 s_2, s_1 s_2 s_0, s_1 s_2 s_0 s_1, s_1 s_2 s_0 s_1 s_2, \dots$

Multiplying the generators cyclically does not introduce braids.

This is not the right question.

Enumerating fully commutative elements

Question: **How many** fully commutative elements are there in $\widetilde{S}_n$, with Coxeter length ℓ ?

In $\widetilde{S}_3$: id , s_0 , s_1 , s_2 , s_0s_1 , s_1s_0 , s_0s_2 , s_2s_0 , s_1s_2 , s_2s_1 , $s_0s_1s_2$, $s_1s_2s_0$, $s_2s_0s_1$, $s_0s_2s_1$, $s_1s_0s_2$, $s_2s_1s_0$, $\dots$

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$$\text{In } \tilde{S}_3: \begin{array}{cccccc} & s_0 & s_0 s_1 & s_0 s_2 & s_0 s_1 s_2 & s_0 s_2 s_1 \\ \text{id}, & s_1 & s_1 s_0 & s_1 s_2 & s_1 s_0 s_2 & s_1 s_2 s_0 & , \dots \\ & s_2 & s_2 s_0 & s_2 s_1 & s_2 s_0 s_1 & s_2 s_1 s_0 \end{array}$$

Question: Determine the coefficient of q^ℓ in the generating function

$$f_n(q) = \sum_{\tilde{w} \in \widetilde{S}_n^{\text{FC}}} q^{\ell(w)}.$$

$$f_3(q) = 1q^0 + 3q^1 + 6q^2 + 6q^3 + \dots$$

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Answer: Consult your friendly computer algebra program.

DdddaaaaAAAAaaaaTTaaaaAA

Brant calls up and says: “Hey Chris, look at this data!”

$$f_3(q) = 1 + 3q + 6q^2 + 6q^3 + 6q^4 + 6q^5 + \cdots$$

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$$f_7(q) = 1 + 7q + 28q^2 + 77q^3 + 161q^4 + 266q^5 + 364q^6 + 427q^7 + 462q^8 + 483q^9 + 490q^{10} + 490q^{11} + 490q^{12} + 490q^{13} + \dots$$

Notice:

- The coefficients eventually repeat.

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Notice:

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Goals: ★ Find a formula for the generating function $f_n(q)$.

★ Understand this periodicity.

Pattern Avoidance Characterization

Key idea: (**Green**, 2002)

$\tilde{w}$ is fully commutative $\iff \tilde{w}$ is 321-avoiding.

Example. $[-4, -1, 1, 14]$ is **NOT** fully commutative because:

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	$\dots w(-4)$	$w(-3)$ $w(-2)$ $w(-1)$ $w(0)$	$w(1)$ $w(2)$ $w(3)$ $w(4)$	$w(5)$ $w(6)$ $w(7)$ $w(8)$	$w(9) \dots$
$\tilde{w}$	$\dots 6$	-8 -5 -3 10	-4 -1 1 14	0 3 5 18	$4 \dots$

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 - ▶ w^0 determines the entries; w determines their order.

Example. For $\tilde{w} = [-11, 20, -3, 4, 11, 0] \in \tilde{S}_6$,

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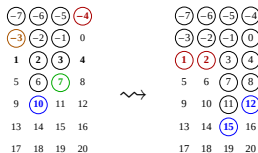
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- ▶ Determine which w^0 are 321-avoiding.
- ▶ Determine the finite w such that $w^0 w$ is still 321-avoiding

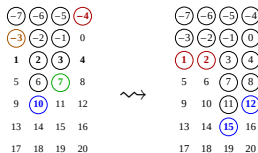
Normalized abacus and 321-avoiding criterion for $\tilde{S}_n/S_n$

We use a *normalized* abacus diagram; shifts all beads so that the first gap is in position $n + 1$; this map is invertible.



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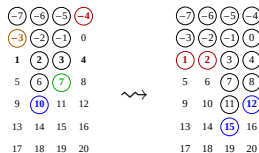


Theorem. (H–J ‘09) Given a normalized abacus for $w^0 \in \widetilde{S}_n/S_n$, where the last bead occurs in position i ,

w^0 is fully commutative $\iff$ lowest beads in runners only occur in $\{1, \dots, n\} \cup \{i - n + 1, \dots, i\}$

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Idea: Lowest beads in runners $\leftrightarrow$ entries in base window.

$w(-n+1)$ $w(-n+2)$... $w(-1)$ $w(0)$	$w(1)$ $w(2)$... $w(n-1)$ $w(n)$	$w(n+1)$ $w(n+2)$... $w(2n-1)$ $w(2n)$
lo lo ... hi hi	lo lo ... hi hi	lo lo ... hi hi
lo lo med hi hi	lo lo med hi hi	lo lo med hi hi

Long versus short elements

Partition $\widetilde{S}_n$ into long and short elements:

Short elements

Lowest bead in position $i \leq 2n$

Finitely many

Hard to count

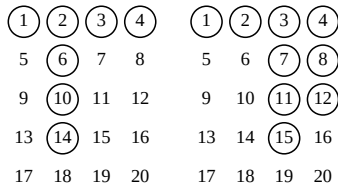
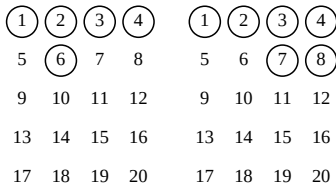
Long elements

Lowest bead in position $i > 2n$

Come in infinite families

Easy to count

Explain the periodicity



Enumerating long elements

For long elements $\tilde{w} \in \widetilde{S}_n$, the base window for w^0 is $[a, a, \dots, a, b, b, \dots, b]$ where $1 \leq a \leq n$, and $n+2 \leq b$.

1	2	3	4
5	6	7	8
9	10	11	12
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- ▶ We can not invert any pairs of a 's, nor any pairs of b 's.
(Would create a 321-pattern with an adjacent window)
- ▶ Only possible to *intersperse* the a 's and the b 's.

How many ways to intersperse (k) a 's and $(n - k)$ b 's? $\binom{n}{k}$

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BUT: We must also keep track of the *length* of these permutations.

This is counted by the q -binomial coefficient: $\begin{bmatrix} n \\ k \end{bmatrix}_q$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q)_n}{(q)_k (q)_{n-k}}, \text{ where } (q)_n = (1-q)(1-q^2) \cdots (1-q^n)$$

Enumerating long elements

After we:

- ▶ Enumerate by length all possible w^0 with (k) a 's and $(n - k)$ b 's.
- ▶ Combine the Coxeter lengths by $\ell(\tilde{w}) = \ell(w^0) + \ell(w)$.

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Then we get:

Theorem. (H-J '09) For a fixed $n \geq 0$, the generating function by length for *long* fully commutative elements $\tilde{w} \in \tilde{S}_n^{FC}$ is

$$\sum q^{\ell(\tilde{w})} = \frac{q^n}{1 - q^n} \sum_{k=1}^{n-1} \begin{bmatrix} n \\ k \end{bmatrix}_q^2.$$

Periodicity of fully commutative elements in $\widetilde{S}_n$

Corollary. (H–J '09) The coefficients of $f_n(q)$ are eventually periodic with period dividing n .

Proof. For i sufficiently large, **all elements** of length i are long. Our generating function is simply **some polynomial** over $(1 - q^n)$:

$$\frac{q^n}{1 - q^n} \sum_{k=1}^{n-1} \left[\begin{matrix} n \\ k \end{matrix} \right]_q^2 = \frac{P(q)}{1 - q^n} = P(q)(1 + q^n + q^{2n} + \cdots)$$

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When n is prime, the period is 1: $a_i = \frac{1}{n} \left(\binom{2n}{n} - 2 \right)$.

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When n is prime, an extra factor of $(1 + q + \cdots + q^{n-1})$ cancels;

$$\frac{1}{1 - q} \left[\frac{q^n}{1 + q + \cdots + q^{n-1}} \sum_{k=1}^{n-1} \left[\begin{matrix} n \\ k \end{matrix} \right]_q^2 \right]$$

As suggested by a referee, we know that $a_i = P(1) = \frac{1}{n} \sum_{k=1}^{n-1} \binom{n}{k}^2$.

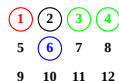
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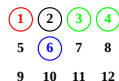


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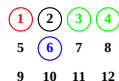


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- ▶ Count $\tilde{w}$ where some a intertwines with some c .
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- ▶ Count $\tilde{w}$ w/o intertwining and 1 descent in the b 's.
 - ▶ Not so hard to determine the acceptable finite permutations w .
 - ▶ Such as $\sum_{M \geq 0} x^{L+M+R} \sum_{\mu=1}^{M-1} \left(\begin{bmatrix} M \\ \mu \end{bmatrix}_q - 1 \right) \begin{bmatrix} L+\mu \\ \mu \end{bmatrix}_q \begin{bmatrix} R+M-\mu \\ M-\mu \end{bmatrix}_q$

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- ▶ Count $\tilde{w}$ w/o intertwining and 2 descents in the b 's.
- ▶ Count $\tilde{w}$ which are finite permutations. (Barcucci et al.)
 - ▶ Solve functional recurrences (Bousquet-Mélou)
 - ▶ Such as $D(x, q, z, s) = N(x, q, z, s) + \frac{xqs}{1-qs} (D(x, q, z, 1) - D(x, q, z, qs)) + xsD(x, q, z, s)$

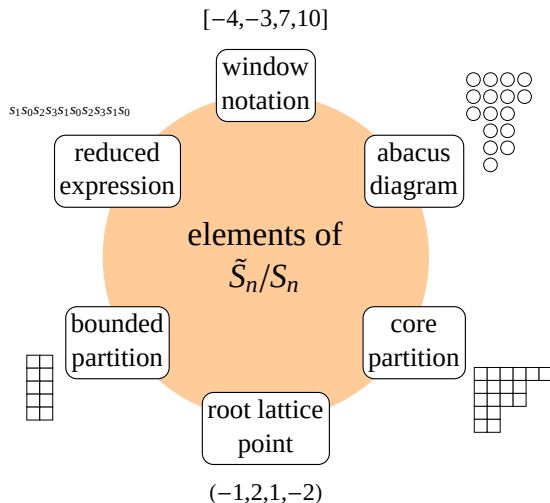
Future Work

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 - ▶ 321-avoiding characterization?

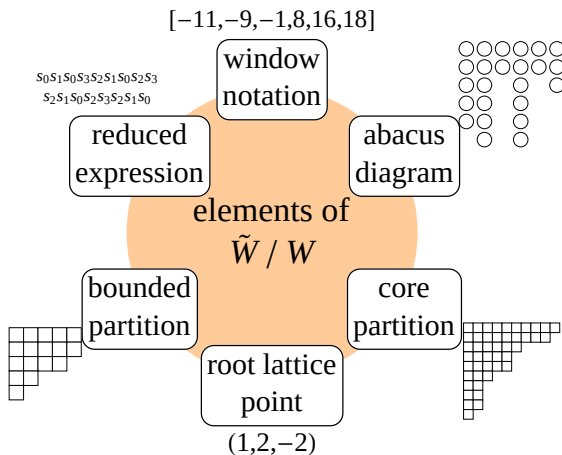
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Combinatorial interpretations of $\tilde{S}_n/S_n$



Combinatorial interpretations of $\tilde{C}/C$, $\tilde{B}/B$, $\tilde{B}/D$, $\tilde{D}/D$



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- ▶ More combinatorial interpretations for $\widetilde{W}/W$
 - ▶ What do you know?

Thank you!

Slides available: people.qc.cuny.edu/chanusa > Talks



Anders Björner and Francesco Brenti.

Combinatorics of Coxeter Groups, Springer, 2005.



Christopher R. H. Hanusa and Brant C. Jones.

The enumeration of fully commutative affine permutations

European Journal of Combinatorics. Vol 31, 1342–1359. (2010)



Christopher R. H. Hanusa and Brant C. Jones.

Abacus models for parabolic quotients of affine Coxeter groups